Name ____ Date ___________ Period _____

Composition to Verify Inverses - Practice

Algebra II | Unit 1 | Lesson 13 Practice | TEK A2.2.D

Objective: I can compose two functions and use both compositions to prove or disprove that they are inverses.

Section 1 — Recall

1. Let f(x) = 4x + 1 and g(x) = (x - 1)/4. Compute f(g(3)) step by step.

g(3) = __________ f(0.5) = __________

2. For the same f and g, compute g(f(2)).

f(2) = __________ g(9) = __________

Section 2 — Apply

3. Test f(x) = 5x + 2 and g(x) = (x - 2)/5 - compute both compositions to decide if they are inverses.

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f(g(x)) = __________ g(f(x)) = __________ Inverses? __________

4. Test f(x) = 3x - 1 and g(x) = 3x + 1 (a "changed sign" pair). Compute f(g(x)) to check.

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f(g(x)) = ________________ Inverses? __________

Section 3 — Challenge

5. A student claims f(x) = 2x + 7 and g(x) = (x - 7)/2 are inverses after checking only f(g(x)) = x. Show g(f(x)) too, and explain whether checking one direction was actually sufficient here or just lucky.

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6. f(x) = 6 - 2x and g(x) = (6 - x)/2. Are these inverses? Show both compositions, and note anything interesting about how f and g compare as expressions.

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f(g(x)) = __________ g(f(x)) = __________ Inverses? __________

Section 4 — Explain It

7. A classmate claims: "If f(g(x)) = x works out at one specific value, like x = 3, that's proof f and g are inverses." Test using f(x) = x + 1 and g(x) = 2x - 4 at x = 3, then at x = 5, to evaluate the claim.

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